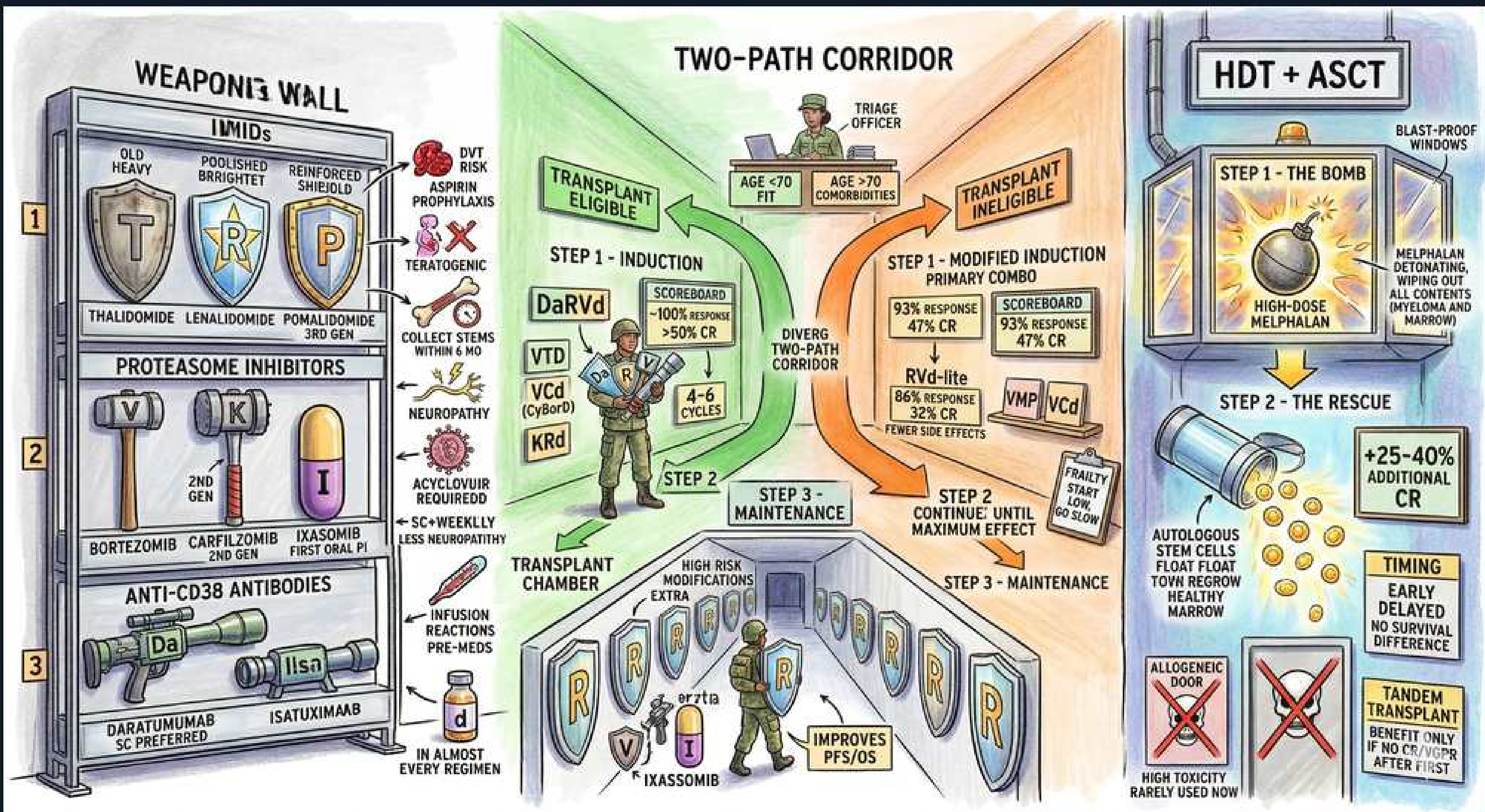
Multiple Myeloma — Frontline Treatment, Induction & ASCT



1. IMiDs (Thal / Len / Pom)

WHAT YOU SEE

A 'WEAPONS WALL — IMiDs' shelf with thalidomide, lenalidomide, pomalidomide shields.

WHAT IT ENCODES

Immunomodulatory drugs (IMiDs) — thalidomide, lenalidomide, pomalidomide — bind cereblon to degrade Ikaros/Aiolos, killing plasma cells and stimulating T/NK immunity. Lenalidomide is the backbone of induction and maintenance. Major toxicities: VTE (needs prophylaxis), myelosuppression (len), and teratogenicity.

BOARD TRAP

All IMiDs are teratogenic (thalidomide phocomelia) and thrombogenic — every patient needs VTE prophylaxis and a pregnancy-prevention program; this is a guaranteed exam point.

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2. Proteasome inhibitors (Bor/Carf/Ixa)

WHAT YOU SEE

A 'PROTEASOME INHIBITORS' rack with bortezomib, carfilzomib, ixazomib hammers.

WHAT IT ENCODES

Proteasome inhibitors (bortezomib, carfilzomib, ixazomib) block the 26S proteasome, causing toxic protein accumulation and apoptosis in plasma cells (which have huge protein output). Bortezomib is renally safe and overcomes t(4;14). Bortezomib causes peripheral neuropathy (reduced with weekly/subcutaneous dosing); carfilzomib causes CARDIOTOXICITY, not neuropathy.

BOARD TRAP

Bortezomib = peripheral neuropathy (give subcut/weekly); carfilzomib = cardiac/hypertensive toxicity (NOT neuropathy) — don't swap their signature side effects. Bortezomib is preferred in renal failure.

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3. Anti-CD38 (Dara / Isa)

WHAT YOU SEE

An 'ANTI-CD38 ANTIBODIES' shelf with daratumumab (SC preferred) and isatuximab.

WHAT IT ENCODES

Anti-CD38 monoclonal antibodies (daratumumab, isatuximab) target CD38 on plasma cells via ADCC/CDC and are now part of frontline quadruplets (Dara-RVd / Dara-VRd). Daratumumab subcutaneous is preferred. CD38 antibodies interfere with blood-bank typing (pan-reactive antibody screen) and SPEP/immunofixation (can mimic residual M-spike).

BOARD TRAP

Anti-CD38 mAbs cause a false-positive indirect Coombs/antibody screen (use DTT-treated cells) and can show up as a faint M-band on immunofixation — alert the blood bank and interpret response carefully.

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4. Transplant-eligible — VRd/Dara-VRd induction

WHAT YOU SEE

A 'TRANSPLANT ELIGIBLE' lane: Step 1 induction with DaRVd / VRd / KRd over 4–6 cycles.

WHAT IT ENCODES

Transplant-eligible patients (typically <70-75, fit) get induction with a triplet (VRd: bortezomib-lenalidomide-dexamethasone) or, increasingly, a quadruplet adding daratumumab (Dara-VRd) for ~4-6 cycles, then proceed to high-dose melphalan with autologous stem-cell transplant.

BOARD TRAP

Eligibility for ASCT is based on FITNESS/comorbidity and physiologic state — age is a guide (~70-75), NOT an absolute cutoff; chronologic age alone shouldn't disqualify a fit patient.

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5. Collect stems before melphalan

WHAT YOU SEE

A note to 'collect stem cells within ~4–6 cycles' before lenalidomide harms harvest.

WHAT IT ENCODES

Because lenalidomide and prolonged alkylator exposure impair stem-cell mobilization, autologous stem cells should be COLLECTED EARLY (after ~4-6 induction cycles) in transplant-eligible patients, then cryopreserved for ASCT.

BOARD TRAP

Prolonged lenalidomide before harvest reduces stem-cell yield — collect stem cells early; don't delay mobilization until after many induction cycles.

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6. Transplant-ineligible — modified RVd-lite

WHAT YOU SEE

A 'TRANSPLANT INELIGIBLE' lane: modified induction (RVd-lite / VRd-lite, VMP, VCd) to maximum effect.

WHAT IT ENCODES

Transplant-ineligible (frail/older) patients receive lower-intensity continuous therapy — Dara-Rd, VRd-lite, or VMP — dosed to maximal response and tolerance rather than a fixed transplant timeline. Geriatric assessment guides dose adjustment to avoid toxicity.

BOARD TRAP

Ineligible does NOT mean undertreated — give continuous, dose-adjusted triplet/quadruplet therapy; the difference is intensity and the absence of high-dose melphalan, not omitting effective drugs.

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7. Step 1 — Melphalan 'the bomb'

WHAT YOU SEE

An 'HDT + ASCT' panel: Step 1 a bomb labeled 'HIGH-DOSE MELPHALAN' detonating in the marrow.

WHAT IT ENCODES

Autologous transplant conditioning is high-dose melphalan 200 mg/m2, an alkylator that ablates marrow (and disease). It deepens and prolongs response. Melphalan dose is reduced (e.g., 140 mg/m2) in renal impairment or frailty. Mucositis and cytopenias are expected.

BOARD TRAP

The conditioning agent for myeloma ASCT is high-dose MELPHALAN (an alkylator) — not total-body irradiation; reduce the dose for renal impairment.

8. Step 2 — Autologous rescue (+25–40% CR)

WHAT YOU SEE

Step 2 'the rescue': previously collected autologous stem cells reinfused, '+25–40% additional CR'.

WHAT IT ENCODES

After melphalan, the patient's own cryopreserved CD34+ stem cells are reinfused to rescue hematopoiesis. ASCT improves depth of response and progression-free survival (deeper CR rates) but, with modern induction, an overall-survival benefit vs delaying transplant to relapse is debated — PFS benefit is the consistent finding.

BOARD TRAP

Up-front ASCT improves PFS (and depth of response) but modern data question a clear OS advantage over collecting and transplanting at relapse — frame the benefit as PFS, with timing individualized.

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9. Maintenance — lenalidomide

WHAT YOU SEE

A 'STEP 3 — MAINTENANCE' chamber after transplant.

WHAT IT ENCODES

Post-ASCT maintenance with lenalidomide continued until progression improves PFS and OS. High-risk patients (del17p, t4;14) often get bortezomib-containing or doublet maintenance. Lenalidomide maintenance carries a small increased risk of secondary primary malignancies — outweighed by survival benefit.

BOARD TRAP

Standard post-transplant maintenance is LENALIDOMIDE to progression; high-risk cytogenetics warrant adding a proteasome inhibitor — single-agent len may be insufficient for del(17p)/t(4;14).

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10. Ixazomib (oral PI) improves PFS

WHAT YOU SEE

An 'IXAZOMIB' note: oral proteasome inhibitor improving PFS in maintenance.

WHAT IT ENCODES

Ixazomib is the ORAL proteasome inhibitor, convenient for all-oral regimens (e.g., IRd: ixazomib-len-dex) and maintenance, with less neuropathy than bortezomib. It can prolong PFS, especially in high-risk patients, though magnitude is modest.

BOARD TRAP

Ixazomib's selling point is ORAL administration with low neuropathy — a convenience/tolerability choice; it is not more potent than IV bortezomib/carfilzomib.

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11. VTE risk / aspirin prophylaxis

WHAT YOU SEE

A 'DVT RISK — ASPIRIN PROPHYLAXIS' icon near the IMiD shelf.

WHAT IT ENCODES

IMiD-based regimens substantially raise venous thromboembolism risk, especially combined with dexamethasone. Low-risk patients get aspirin prophylaxis; higher-risk (prior VTE, immobility, additional risk factors) get therapeutic anticoagulation (LMWH/DOAC) throughout IMiD therapy.

BOARD TRAP

Every patient on an IMiD needs VTE prophylaxis — aspirin for low risk, anticoagulation for high risk; omitting it is a tested error.

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12. Neuropathy (bortezomib/thalidomide)

WHAT YOU SEE

A 'NEUROPATHY' warning by the proteasome-inhibitor rack.

WHAT IT ENCODES

Peripheral sensory neuropathy is the dose-limiting toxicity of bortezomib and thalidomide. Mitigation: switch bortezomib to once-weekly subcutaneous dosing, dose-reduce, and monitor. Carfilzomib spares nerves but is cardiotoxic; ixazomib has minimal neuropathy.

BOARD TRAP

To reduce bortezomib neuropathy switch from IV twice-weekly to SUBCUTANEOUS once-weekly — and remember carfilzomib avoids neuropathy at the cost of cardiac toxicity.

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13. Acyclovir prophylaxis (PI/HZ)

WHAT YOU SEE

An 'ACYCLOVIR REQUIRED' note attached to the proteasome inhibitors.

WHAT IT ENCODES

Proteasome inhibitors (and anti-CD38, transplant) markedly increase herpes zoster reactivation risk, so acyclovir/valacyclovir prophylaxis is mandatory during bortezomib/carfilzomib therapy. PJP prophylaxis is also commonly used with high-dose steroids.

BOARD TRAP

Patients on bortezomib/carfilzomib REQUIRE antiviral (acyclovir) prophylaxis against VZV reactivation — a frequently tested supportive-care detail.

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14. Allogeneic — rarely used

WHAT YOU SEE

An 'ALLOGENEIC DOOR' marked high-toxicity, rarely used; a 'TANDEM TRANSPLANT' note for high risk.

WHAT IT ENCODES

Allogeneic transplant offers a graft-versus-myeloma effect but carries high transplant-related mortality and GVHD, so it is reserved for select young high-risk/relapsed patients on trials. Tandem (two sequential) autologous transplants may benefit very-high-risk cytogenetics.

BOARD TRAP

Allogeneic transplant is NOT standard in myeloma (high TRM, GVHD) — the standard is AUTOLOGOUS; allo is investigational/select high-risk only.

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15. Two-path corridor (eligible vs not)

WHAT YOU SEE

A 'TWO-PATH CORRIDOR' diverging into transplant-eligible vs ineligible after a triage officer assesses age/fitness.

WHAT IT ENCODES

The frontline algorithm forks at transplant eligibility, determined by fitness, comorbidity, organ function, and patient preference. Both arms start with proteasome-inhibitor + IMiD + steroid (± anti-CD38); only the eligible arm proceeds to high-dose melphalan/ASCT. Both continue maintenance.

BOARD TRAP

The first decision in newly diagnosed myeloma is transplant ELIGIBILITY (fitness, not just age) — it determines whether high-dose melphalan/ASCT is part of the plan, but induction backbone is similar.